European Journal of Cancer 241 (2026) 116788

Contents lists available at ScienceDirect

European Journal of Cancer

journal homepage: www.ejcancer.com

Original Research

Performance of FIT-based colorectal cancer screening in Denmark from 2014 to 2022

Tina Bech Olesen [a,1], Signe Bülow Therkildsen [a,b,*,1], Petra Hall Viborg [c], Nina Brander [d], Lennart Friis-Hansen [e,f], Mette Høyrup [c], Henry Jensen [c], Sanne Kjær-Frifeldt [g], Lasse Pedersen [h], Mikkel Sjöström [i], Bo Søndergaard [j], Morten Rasmussen [k,l], Berit Andersen [a,b]

[a] University Research Clinic for Cancer Screening (UNICCA), Department of Public Health Programmes, Randers Regional Hospital, Randers, Denmark
[b] Department of Clinical Medicine, Aarhus University, Aarhus, Denmark
[c] The Danish Healthcare Quality Institute (DHQI), Aarhus and Copenhagen, Denmark
[d] Department of Gastrointestinal Surgery, Slagelse Hospital, Denmark
[e] Department of Clinical Biochemistry, Odense University Hospital, Odense, Denmark
[f] Department of Clinical Medicine, University of Southern Denmark, Odense, Denmark
[g] Department of Clinical Pathology, Lillebælt Hospital, Vejle, Denmark
[h] Department of Gastrointestinal Surgery, Aalborg University Hospital, Denmark
[i] Department of Gastrointestinal Surgery, Vejle University Hospital, University of Southern Denmark, Denmark
[j] Gastrounit, Copenhagen University Hospital - Amager and Hvidovre, Hvidovre, Denmark
[k] The Colorectal Cancer Screening Programme, The Capital Region, Copenhagen, Denmark
[l] Digestive Disease Center, Bispebjerg University Hospital, Copenhagen, Denmark

ARTICLE INFO

Keywords:
Epidemiology
Colorectal cancer screening
Participation
Faecal immunochemical test (FIT)
Compliance
Colonoscopy
Adenoma detection rate (ADR)
Post-colonoscopy colorectal cancer (PCCRC)
Surveillance

ABSTRACT

Background: Faecal immunochemical test (FIT)-based colorectal cancer (CRC) screening was gradually implemented in Denmark from March 2014 to December 2017. From January 2018, the FIT screening became biennial and is offered to all citizens aged 50–74 years. Citizens with a positive FIT (≥20 μg haemoglobin/g faeces), are scheduled for colonoscopy within 14 days. We examined indicators of quality in FIT-based CRC screening over time and compared with European guidelines. Furthermore, the overall CRC incidence and mortality before, during and after the implementation of CRC screening were examined.

Methods: Nation-wide register data on CRC screening in Denmark were used to assess screening performance during prevalence screening (2014–2017) and incidence screening (2018–2022). For both prevalence and incidence screening we compared results with quality indicators from the European guidelines. Aggregated data on CRC incidence and mortality were obtained from surveillance data.

Results: During prevalence and incidence screening, participation in CRC screening was 61.7% and 60.5%, the proportion of positive FITs were 6.9% and 4.7%, respectively. The compliance to follow-up colonoscopy was approximately 90%. The adenoma detection rate increased slightly over time, while number of screen-detected CRC decreased over time. The 3-year post-colonoscopy CRC rate fluctuated around 7.1–8.9% and decreased to 6.8% in 2021. A CRC incidence peak was observed after the introduction of screening followed by a decrease, while mortality continued to decline.

Conclusion: The performance of FIT-based CRC screening in Denmark is within the acceptable levels defined by the European guidelines; however, improvements are needed, particularly an increase in participation.

* Corresponding author at: University Research Clinic for Cancer Screening (UNICCA), Department of Public Health Programmes, Randers Regional Hospital, Randers, Denmark.
E-mail address: sbt@clin.au.dk (S.B. Therkildsen).
[1] Shared first authorship.

https://doi.org/10.1016/j.ejca.2026.116788
Received 12 December 2025; Received in revised form 23 April 2026; Accepted 3 May 2026
Available online 12 May 2026
0959-8049/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Colorectal cancer (CRC) is the third most common cancer worldwide [1]. In Denmark, CRC accounts for approximately 4500 new patients and 1671 deaths annually [2].

Since 2003, CRC screening has been recommended by the European Council [3] and this has led to the implementation of CRC screening in many European countries [4]. The aim of CRC screening is to reduce the incidence, morbidity and mortality from CRC by removing adenomas and detecting CRC at earlier stages [5].

To improve the quality and effectiveness of CRC screening programmes in Europe, European guidelines for quality assurance and indicators of quality in CRC screening and diagnosis have been developed by a group of experts appointed by the International Agency for Research on Cancer (IARC), the first edition published in 2012 [6]. Moreover, currently, new guidelines are being developed by the European Commission Initiative on Colorectal Cancer (ECICC) summarising the available evidence on CRC screening age, frequency, tests and invitation strategies. These new guidelines recommend organised, population-based CRC screening for all adults aged 50–74 years old every two years using faecal immunochemical test (FIT) [7].

In countries with a high coverage and uptake of CRC screening, earlier detection leads to an initial increase in the CRC incidence, followed by a decrease in the incidence over time [4,8]. Some studies have also shown a reduction in mortality due to CRC after the introduction of FIT-based CRC screening [9,10], although other studies found the evidence to be inconclusive [11–13].

In this study, we examined indicators of quality in the FIT-based CRC screening programme in Denmark over time and in comparison, with the European guidelines for CRC screening. Furthermore, we examined the overall incidence and mortality of CRC in the entire population before, during and after the implementation of CRC screening in Denmark.

2. Material and methods

2.1. Setting

This study was set in Denmark with a population of approximately 6.0 million inhabitants [14]. All Danish citizens have access to universal, tax-supported health care, including participation in national cancer screening programmes and any subsequent diagnostic procedures and treatment. Extensive administrative and health data are recorded in nation-wide population-based registries, and these data can be linked using a unique personal identifier, which is assigned to all citizens at birth or at immigration [15].

2.2. The colorectal cancer screening programme

The CRC screening programme was implemented in Denmark from March 2014 offering FIT screening to all citizens aged 50–74 years [16]. The programme was gradually implemented over a 4-year period from March 2014 to December 2017 (prevalence screening) and as of January 2018, the screening programme was fully implemented nation-wide. From this time, the screening became biennial (incidence screening). During the implementation period (prevalence screening) all eligible citizens were randomly invited once in an order based on their birth month and then re invited again in incidence screening, resulting in two to four years between the first and the second invitation to FIT screening. Onwards citizens were invited again two years since their last screening or invitation (if they did not participate). When citizens turn 50 years, they will enter the programme and receive their first invitation [17]. The invitation for CRC screening includes a FIT kit (OC Sensor; Eiken Chemical Company, Tokyo, Japan) for obtaining a stool sample at home and a pre-stamped, pre-addressed return envelope. A single-sample FIT is used as the screening procedure. The test can detect low amounts of blood in stool samples, which potentially could be caused by bleeding lesions from precancerous adenomas or CRC [18,19]. At the time of the study, one reminder was sent via digital mail, if a stool sample was not returned within 45 days. Citizens excused from digital mail receive a postal letter for invitations and reminders.

Following a positive FIT (≥20 μg haemoglobin/g faeces), citizens are scheduled for colonoscopy within 14 days to support timely follow-up. Screening participants with a positive FIT receive bowel preparation by mail or at the local pharmacy to be initiated at home before arriving at the hospital; however, there is no consultation with a healthcare professional before making the decision to undergo colonoscopy.

Based on the follow-up assessment, colonoscopy outcomes in the period 2014–2022 were categorised into: clean colon, low-risk adenomas (1–2 adenomas <10 mm), intermediate risk adenomas (3–4 adenomas <10 mm or ≥1 adenoma 10–20mm), high-risk adenomas (≥5 adenomas <10 mm or ≥1 adenomas ≥20 mm) [20] or CRC. Guidelines for follow-up after any significant finding is described elsewhere [21], but in brief, if intermediate- or high-risk adenomas were detected, the individuals were referred to a follow-up programme with colonoscopy after three and one year(s), respectively, and if low-risk adenomas were detected the participants were invited to FIT-screening again after two years. Individuals with a clean colon were invited to FIT-screening again after eight years. New guidelines for the categorisation of CRC screening outcome were introduced in 2023; however, in this study the existing guideline from 2016 was used due to the study period (2014–2022).

Data related to the quality of CRC screening in Denmark are available in the Danish Colorectal Cancer Screening Database (DCCSD) [16], which was established to monitor the performance of the screening programme and publishes annual reports.

2.3. Data sources

We used individual data from DCCSD [16], the Danish National Patient Registry [22] and the Danish Civil Registration System [23]. Information on citizens invited for CRC screening, participation etc. was obtained from DCCSD, information on colonoscopies, sigmoidoscopies, and CT-colonografies was collected from the Danish National Patient Registry and finally, information on municipality, emigration and death was collected from the Danish Civil Registration System. Aggregated data on CRC incidence and mortality was obtained from national surveillance data from NORDCAN [2].

2.4. Study population

The study population as identified in DCCSD comprised all individuals aged 50–74 years invited to participate in CRC screening from the initiation of the screening programme on 3 March 2014–31 December 2022. Aggregated data on incidence and mortality comprised all individuals aged ≥ 50 years in the entire population.

2.5. Outcomes of interest

We report the following outcomes based on data from DCCSD [16]:

- The number and proportion of citizens who participated in CRC screening by returning a FIT within 90 days of invitation or reminder among all citizens invited.
- The number and proportion of positive (a sample with a value ≥20 μg haemoglobin/g faeces, in the first returned test), negative (a sample with a value <20 μg haemoglobin/g faeces) and inadequate FIT (sample not suitable for analysis) in the first returned FIT sample after an invitation.
- The number and proportion of total positive FIT (participants with a positive FIT and participants with a positive FIT following an inadequate first FIT).

- The compliance to diagnostic assessment of the colon (colonoscopy, sigmoidoscopy, CT-colonography, or capsule endoscopy) within 60 days since a positive FIT among participants.
- The number and proportion of complete colonoscopies among colonoscopies with known completeness (cecum intubation rate) during follow-up colon assessment following a positive FIT.
- The number and proportion of adenomas (adenoma detection rate (ADR)) and sessile serrated lesions (SSL) among all follow-up procedures and classification of adenomas (low-risk, intermediate risk, high-risk, unknown).
- The number and proportion of screen-detected CRC among all follow-up procedures.
- The number of screen-detected CRC per 100,000 screening invitees and 95% confidence interval (CI) and per 100,000 screening participants and 95% CI.
- The number of interval CRC per 10,000 participants and 95% CI within two years since a negative FIT.
- The proportion of post-colonoscopy CRC (PCCRC) within three years since a screening-derived colonoscopy after a positive FIT without detection of CRC.

Additionally, we compared participation, the proportion of inadequate FITs, compliance to colonoscopy after a positive FIT, and the proportion of complete colonoscopies within the CRC screening programme in Denmark with the acceptable and desirable levels of these measures as stated by the European guidelines for quality assurance in CRC screening [6].

Finally, using aggregated data from NORDCAN [2], we assessed incidence and mortality of CRC in the entire population before the introduction of screening, during the implementation of screening and after the implementation of screening.

The abovementioned outcomes of interest are defined in more detail in Supplementary Table 1.

2.6. Statistical analyses

Descriptive analyses based on individually linked data were performed for each outcome of interest by calculating numbers and proportions. The number of screen-detected CRC and 95% CI were assessed per 100,000 invitees and per 100,000 participants. The number of interval cancers and 95% CI were assessed per 10,000 participants. The 3-year PCCRC proportion was calculated according to the method recommended by World Endoscopy Organisation (WEO) as follows: PCCRC rate = (number of false negative colonoscopies/(number of false negative colonoscopies + number of true positive colonoscopies)) * 100 [24]. A false negative colonoscopy is defined as $\geq$ 1 colonoscopy after colorectal cancer screening within 6–36 months prior to a diagnosis of CRC and a true positive colonoscopy is defined as $\geq$ 1 colonoscopy after colorectal cancer screening within 0–6 months prior to a diagnosis of CRC.

The analyses were stratified by year of invitation and cancer site (colon, rectum, unknown). In supplementary analyses, the analyses were also stratified by age and sex. Moreover, the analyses were assessed separately for prevalence screening (2014–2017) and incidence screening (2018–2022), given that the risk of a positive FIT and the risk of CRC differ between the two populations.

All analyses using individual-level data were performed at the Danish Healthcare Quality Institute as part of the annual reporting of quality data. The analyses were carried out using SAS Enterprise Guide (version 9.2).

At an aggregated level, the age-standardised (Nordic) incidence and mortality rate and 95% CI of CRC were assessed per 100,000 person-years from 2008 to 2022 among individuals aged $\geq$ 50 years old in the entire population. The analyses were also stratified by age and sex. Moreover, we assessed the estimated annual percentage change (EAPC) of the age-standardised incidence and mortality rate of CRC from 2008 to 2022 based on the available data at NORDCAN (EAPC is only available in 5, 10, 15, 20 years intervals). The EAPC describes the magnitude of the change in trend and is calculated as the average annual rate of change in the ASR over the time period selected. EAPC is computed by fitting a regression line to the natural logarithm of the ASR using calendar year as a regressor variable [2]. Additionally, we examined the crude incidence and mortality and 95% CI of CRC.

2.7. Ethical considerations

We used data from the DCCSD annual reports and publicly available surveillance data from NORDCAN. Therefore, no additional registration nor ethical approval was required for this study. Patient consent is not required by Danish law for register-based studies.

3. Results

3.1. Participation, proportion of inadequate FIT and positive FIT and compliance to follow-up

Altogether, 2,032,507 invitations for CRC screening were sent from 2014 to 2017 (prevalence screening) and 4,292,805 invitations from 2018 to 2022 (incidence screening).

During prevalence and incidence screening, 61.7% and 60.5% participated in screening within 90 days since invitation, the proportion of inadequate FITs were low (0.4% and 0.3%), whilst the proportion of positive FIT was 6.9% during prevalence screening and 4.7% during incidence screening. The compliance to follow-up within 60 days since a positive FIT was approximately 90% throughout the study period, and the proportion of complete colonoscopies among coloscopies with known completeness was 91.9% during prevalence screening and 92.2% during incidence screening (Table 1).

Compared to the European guidelines for CRC screening, the overall participation, proportion of inadequate FIT, compliance to follow-up and proportion of complete colonoscopies were within the acceptable levels; however, only the proportion of inadequate FIT was within the desirable level (Table 2).

Throughout the study period, the participation in CRC screening was higher among women than among men (65.9% vs 57.5% during prevalence screening and 65.2% vs 55.7% during incidence screening). Conversely, the proportion of positive FIT was higher among men than among women (8.3% vs 5.6% during prevalence screening and 5.5% vs 4.0% during incidence screening). The compliance to follow-up was similar among men and women (Supplementary Table 2 and Supplementary Fig. 1).

The participation in CRC screening increased with increasing age during both incidence and prevalence screening, though with a slight drop in the participation among the oldest age group. The proportion of positive FIT also increased with increasing age during both incidence and prevalence screening (Supplementary Table 3).

3.2. Outcomes of colonoscopies

During prevalence screening, ADR was 56.8% and, during incidence screening, ADR increased slightly over time (2018: 54.7%; 2019: 55.1%; 2020: 55.4%; 2021: 55.8%; 2022: 57.0%). Moreover, there was a shift towards a higher proportion of low-risk adenomas: during prevalence screening, 36.7% of adenomas were low-risk, 31.0% were intermediate risk, and 25.9% were high-risk, whilst during incidence screening, 45.4% of adenomas were low-risk, 28.3% were intermediate risk and 20.6% were high-risk (Table 1).

During prevalence screening, 4577 CRC (5.9% of all follow-up procedures) were detected and 4318 CRC (3.9% of all follow-up procedures) were detected during incidence screening. The majority of CRC were colon cancers (69.9% during prevalence screening and 63.9% during incidence screening) (Table 1). During prevalence screening, the

Table 1
Participation, proportion of positive FIT, compliance to follow-up and number of screen-detected adenomas and colorectal cancers in the FIT-based colorectal cancer screening programme in Denmark, 2014–2022.

	Prevalence	Incidence					
Year	**2014–2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**Total**
Invited N	**2,032,507**	**877,486**	**865,999**	**825,100**	**852,291**	**871,929**	**4,292,805**
Participants n (%)	**1,254,179 (61.7)**	**532,994 (60.7)**	**523,106 (60.4)**	**497,567 (60.3)**	**520,706 (61.1)**	**521,365 (59.8)**	**2,595,738 (60.5)**
Negative FIT Inadequate FIT	1,163,438 (92.8)	505,080 (94.8)	495,091 (94.6)	471,817 (94.8)	495,755 (95.2)	499,246 (95.8)	2,466,989 (95.0)
Positive FIT	4944 (0.4) 85,797 (6.8)	1420 (0.3) 26,494 (5.0)	1695 (0.3) 26,320 (5.0)	1508 (0.3) 24,242 (4.9)	947 (0.18) 24,004 (4.6)	822 (0.2) 21,297 (4.1)	6392 (0.3) 122,357 (4.7)
Total positive FIT n (%)*	**86,242 (6.9)**	**26,596 (5.0)**	**26,417 (5.1)**	**24,324 (4.9)**	**24,045 (4.6)**	**21,331 (4.1)**	**122,731 (4.7)**
Compliance to follow-up n (%)	**77,199 (89.5)**	**23,983 (90.2)**	**23,702 (89.7)**	**21,560 (88.6)**	**21,454 (89.2)**	**19,228 (90.1)**	**109,927 (89.6)**
Compliance to follow-up (only colonoscopies) n (%)	**76,915 (89.2)**	**23,910 (89.9)**	**23,605 (89.4)**	**21,433 (88.1)**	**21,139 (87.9)**	**18,912 (88.7)**	**108,999 (88.8)**
Colonoscopies with known completeness	72,453 (94.2)	23,015 (96.3)	22,728 (96.3)	20,505 (95.7)	20,385 (96.4)	18,251 (96.5)	104,884 (96.2)
Complete colonoscopies out of known completeness	66,564 (91.9)	21,334 (92.7)	21,041 (92.6)	18,881 (92.1)	18,670 (91.6)	16,751 (91.8)	96,677 (92.2)
Procedures with adenomas detected out of all follow-up procedures n (%)	**43,898 (56.9)**	**13,122 (54.7)**	**13,069 (55.1)**	**11,938 (55.4)**	**11,979 (55.8)**	**10,964 (57.0)**	**61,072 (55.6)**
Low-risk Intermediate-risk	16,107 (36.7)	5699 (43.4)	5576 (42.7)	5572 (46.7)	5672 (47.4)	5232 (47.7)	27,751 (45.4)
High-risk Unknown	13,606 (31.0) 11,353 (25.9) 2832 (6.5)	3848 (29.3) 2892 (22.0) 683 (5.2)	3813 (29.2) 2777 (21.3) 903 (6.9)	3318 (27.8) 2314 (19.4) 734 (6.2)	3312 (27.7) 2385 (19.9) 610 (5.1)	3015 (27.5) 2187 (20.0) 530 (4.8)	17,306 (28.3) 12,555 (20.6) 3460 (5.7)
Screen-detected colorectal cancers out of all follow-up procedures n (%)	**4577 (5.9)**	**1020 (4.3)**	**986 (4.2)**	**744 (3.5)**	**851 (4.0)**	**717 (3.7)**	**4318 (3.9)**
Colon cancers Rectum cancers Unknown	3201 (69.9) 1280 (28.0) 96 (2.1)	653 (64.2) 340 (33.3) 27 (2.65)	639 (64.8) 318 (32.3) 31 (3.1)	482 (64.8) 237 (31.9) 25 (3.4)	537 (63.1) 293 (34.4) 21 (2.5)	446 (62.2) 223 (31.1) 48 (6.7) **	2757 (63.9) 1411 (32.6) 152 (3.5)

Abbreviations: FIT = faecal immunochemical test.
* Participants with a positive FIT following an inadequate first FIT are included.
** A large proportion of unknown because of a delay in the data.

Table 2
Performance of the FIT-based colorectal cancer screening programme in Denmark compared to the European Guidelines for colorectal cancer screening.

Quality indicator	Colorectal cancer screening in Denmark						European Guidelines	
	Prevalence (2014–2017)			Incidence (2018–2022)			Acceptable level	Desirable level
	n	N	%	n	N	%	%	%
Participation	1,254,179	2,032,507	61.7	2,595,738	4,292,805	60.5	> 45	> 65
Proportion of inadequate FIT	4944	1,254,179	0.4	6392	2,595,738	0.3	< 3	< 1
Compliance to colonoscopy after a positive FIT*	76,915	86,242	89.2	108,999	122,731	88.8	85	> 90
Proportion of complete colonoscopies**	66,564	72,453	91.9	96,677	104,884	92.2	> 90	> 95

Abbreviations: FIT = faecal immunochemical test.
* Diagnostic assessment of the colon within 60 days since a positive FIT among participants.
** Among colonoscopies with known completeness.

number of CRC per 100,000 invitees decreased from 234.3 (95% CI: 219.6–250.0) in 2014 to 217.5 (95% CI: 206.0–229.7) in 2017 and this tendency was even more pronounced during incidence screening, where the number of CRC per 100,000 invitees decreased from 116.2 (95% CI: 109.3–123.6) in 2018 to 82.3 (95% CI: 76.5–88.5) in 2022. The number of CRC per 100,000 participants was stable during prevalence screening at around 360 CRC per 100,000 participants, whilst the number of CRC per 100,000 participants subsequently decreased from 191.4 (95% CI: 179.9–203.5) in 2018 to 137.5 (95% CI: 127.8–148.0) in 2022 (Table 3).

The number of CRC per 100,000 invitees and per 100,000 participants was higher among men than women. Thus, during prevalence screening, the number of CRC per 100,000 participants was around 470 and 270 for men and women, respectively. During incidence screening, the number of CRC per 100,000 participants decreased from 229.6 (95% CI: 211.3–250.0) to 165.7 (95% CI: 150.1–183.0) among men and from 159.4 (95% CI: 145.5–174.6) to 114.1 (95% CI: 102.3–127.2) among women (Supplementary Table 4).

3.3. Interval and post-colonoscopy CRC

Interval cancers detected within two years of a negative FIT fluctuated around nine per 10,000 participants during both prevalence and incidence screening. The 3-year PCCRC rate fluctuated around 7.1–8.9% throughout the study period and decreased slightly to 6.8% in 2021 (Table 4).

3.4. Incidence and mortality of CRC in the entire population

The overall incidence and mortality of CRC in the entire population (screen- and non-screen-detected CRC) is shown in Figure 1, with higher rates among men than among women (Supplementary Fig. 3). A plateau in the age-standardised (Nordic) incidence of CRC was seen before the introduction of screening followed by a large increase in the incidence of CRC during the implementation of CRC screening and thereafter a gradual decline in the incidence of CRC over time. In particular, a peak

Table 3
Number of screen-detected colorectal cancers per 100,000 invitees and per 100,000 participants in the FIT-based colorectal cancer screening programme in Denmark, 2014–2022.

	Prevalence				Incidence				
Year	**2014**	**2015**	**2016**	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**
Invited N	390,508	495,642	594,843	594,843	877,486	865,999	825,100	852,291	871,929
Participants N	249,196	315,237	335,985	353,788	532,994	523,106	497,567	520,706	521,365
Cancers N	915	1125	1243	1294	1020	986	744	851	717
Colorectal cancer rate pr. 100,000 invited (95%CI)									
All CRC	**234.3 (219.6–250.0)**	**226.9 (214.1–240.6)**	**225.4 (213.2–238.3)**	**217.5 (206.0–229.7)**	**116.2 (109.3–123.6)**	**113.9 (107.9–121.2)**	**90.2(83.9–96.9)**	**99.8(93.4–106.8)**	**82.3(76.5–88.5)**
Colon cancer	166.5 (154.1–179.8)	138.8 (128.8–149.6)	157.0(146.9–167.8)	152.5(142.9–162.7)	74.4(68.9–80.4)	73.8(68.3–79.7)	77.4(71.7–83.7)	63(57.9–68.6)	51.2(46.7–56.2)
Rectum cancer	62.7 (55.4–71.1)	63.8 (57.1–71.2)	64.0(57.7–71.0)	61.5(55.5–68.2)	38.7(34.8–43.1)	36.7(32.9–40.9)	28.7(25.3–32.6)	34.4(30.7–38.5)	25.6(22.4–29.2)
Unknown	5.12 (3.30–7.94)	6.26 (4.4–8.9)	4.35(2.92–6.49)	3.5(2.3–5.4)	3.08(2.1–4.9)	3.35(2.3–4.8)	3.03(2.1–4.5)	2.46(1.6–3.8)	5.5(4.2–7.3)*
Colorectal cancer rate pr. 100,000 participants (95%CI)									
All CRC	**367.2(344.1–391.7)**	**356.9(336.6–378.4)**	**369.9 (349.9–391.1)**	**365.8 (346.3–386.2)**	**191.4 (179.9–203.5)**	**188.5 (177.1–200.6)**	**149.5 (139.2–160.7)**	**163.3 (152.8–174.8)**	**137.5 (127.8–148.0)**
Colon cancer	260.8(241.5–281.7)	218.2(202.5–235.2)	257.8(241.1–275.5)	256.4(240.2–273.6)	122.5(113.4–132.3)	122.2(113.0–132.0)	96.9(88.6–105.9)	103.1(94.8–112.2)	85.5(78.0–93.9)
Rectum cancer	98.3(86.7–111.4)	100.2(89.7–111.9)	105.1(94.7–116.6)	103.5(93.4–114.6)	63.8(57.4–79.9)	60.8(54.5–67.9)	47.6(41.9–54.1)	56.3(50.2–63.1)	42.8(37.5–48.8)
Unknown	8.03(5.2–12.4)	9.8(6.9–14.0)	7.1(4.8–10.7)	5.9(3.9–9.1)	5.07(3.5–7.4)	5.54(3.9–8.0)	5.02(3.4–7.4)	4.03(2.6–6.2)	9.2(6.9–12.2)*

Abbreviations: FIT = faecal immunochemical test; CRC = colorectal cancer.
* A large proportion of unknown because of a delay in the data.

in the incidence of colon cancer was seen during prevalence screening. There was an overall reduction in the incidence of CRC (EAPC: −1.23; 95% CI: −2.32; −0.12) from 2008 to 2022, which thus includes both the peak during prevalence screening and the subsequent decline during incidence screening. During incidence screening, a larger reduction in the incidence of CRC was observed; however, with a wider confidence interval (EAPC: −3.66; 95% CI: −5.48; −1.80).

A slight decline in the age-standardised (Nordic) mortality of CRC was observed already before the introduction of screening (EAPC −3.34; 95% CI: −4.63; −2.04 from 2009 to 2013) (data not shown) and this decline continued with the implementation of screening. During the entire study period (2008–2022), there was an overall decrease in the mortality of CRC (EAPC: −3.28; 95% CI: −3.58; −2.97) (Figure 1 and Supplementary Table 5). Similar patterns were observed when examining the crude incidence and mortality of CRC (Supplementary Fig. 2).

4. Discussion

4.1. Main findings

In this nation-wide, population-based study, we examined the performance of the CRC screening programme in Denmark from 2014 to 2022. Approximately 60% of invitees participated in CRC screening, the proportion of positive FIT was 6.9% during prevalence screening and 4.7% during incidence screening, whilst the compliance to follow-up was a little less than 90% throughout the study period. All measures of CRC screening were within the acceptable levels of the European guidelines for CRC screening. ADR increased slightly over time. The 3-year PCCRC-rate fluctuated around 7.1–8.9 during the study period and decreased slightly to 6.8 in 2021. The number of screen-detected CRC decreased over time. A peak in the incidence of CRC in the entire population was observed after the introduction of screening followed by a subsequent decline. The mortality of CRC in the population declined already before the introduction of screening and this decline continued with the implementation of screening.

4.2. Interpretation of results and comparison with previous studies

The new ECICC guidelines recommend organised, population-based CRC screening for all adults aged 50–74 years old every two years using FIT and recommend sending the invitation letter together with a self-sampling FIT kit to the eligible adults [7]. The CRC screening programme in Denmark is thus in agreement with the ECICC guidelines.

The overall participation in CRC screening was approximately 60% throughout the study period. This estimate is in accordance with a systematic review by Baile-Maxía et al reporting a pooled participation in FIT-based screening among individuals aged 50–74 years old of approximately 55% across Europe, USA, Australia and South Korea [25]. The participation is thus within the acceptable levels, though does not exceed the desirable level (>65%) as defined by the European guidelines. In line with previous studies [26–28], we found a lower participation in CRC screening among men. Furthermore, previous studies have shown a lower participation among younger individuals, immigrants, individuals living alone and individuals with a low income and a low educational level [26–28]. Thus, CRC screening programmes may improve by introducing interventions to increase participation in the general populations as well as among groups of individuals with the lowest participation. Recently, the Danish Cancer Society has launched a campaign to encourage, in particular, younger men in Denmark to take part in CRC screening [29]. Moreover, a pre-notification and an extra reminder have been added to the CRC screening programme in Denmark since November 2024 as this has been shown to increase participation [30]. Possibly, these interventions could increase the participation in CRC screening. Additionally, a randomised controlled trial by Robb et al [31] recently showed that adding a single sentence suggesting a deadline for FIT return in the invitation letter led to a more timely return of

Table 4
Interval and post-colonoscopy colorectal cancers in the FIT-based colorectal cancer screening programme in Denmark, 2014–2021.

	Prevalence screening				Incidence screening			
Year	2014	2015	2016	2017	2018	2019	2020	2021
INTERVAL COLORECTAL CANCER								
Number of interval CRC after a negative FIT per 10,000 participants* (95%CI)	9.4 (8.2.10.8)	9.4 (8.3–10.6)	10.0 (8.9–11.1)	8.8 (7.9–9.9)	9.2 (8.4–10.1)	9.2 (8.4–10.1)	8.5 (7.7–9.4)	NA
POST COLONOSCOPY COLORECTAL CANCER								
Number of false negative colonoscopies**	62	95	118	117	93	116	83	75
Number of true positive colonoscopies***	721	1194	1285	1446	1220	1181	991	1035
Three-year post-colonoscopy CRC rat% ****	7.9	7.4	8.4	7.5	7.1	8.9	7.7	6.8

Abbreviations: FIT = faecal immunochemical test; CRC = colorectal cancer; CI = confidence interval.
* Participants with a negative FIT who were diagnosed with CRC within two years since the negative FIT and/or before the next invitation to FIT screening.
** False negative defined as ≥ 1 colonoscopy after colorectal cancer screening within 6–36 months prior to a diagnosis of CRC.
*** True positive is defined as ≥ 1 colonoscopy after colorectal cancer screening within 0–6 months prior to a diagnosis of CRC.
**** PCCRC = (number of false negative colonoscopies/(number of false negative colonoscopies + number of true positive colonoscopies)) * 100.

the FIT sample, and such an intervention may be another possibility to improve the programme with low costs. Moreover, we have recently shown that offering FIT-screening to women overdue with CRC screening when they attend breast cancer screening was a feasible intervention, resulting in an increase in participation in CRC screening [32]. These studies underline further opportunities for improvements to the CRC screening programme in Denmark to increase participation.

The compliance to follow-up within 60 days since a positive FIT was almost 90% throughout the study period. This exceeds the acceptable level defined by the European guidelines. The systematic review by Baile-Maxía et al showed a compliance to follow-up of 87% [25]. A high overall compliance to follow-up is thus observed in Denmark, though still 10% of screening participants with a positive FIT are not followed up within the intended timeframe. The reasons for non-compliance in our data are unknown; however, the strategy to mail out invitations with a test-kit directly to all citizens eligible for screening could possibly prompt some citizens to participate even though they are unable to undergo colonoscopy or are not eligible to participate in FIT screening or undergo colonoscopy due to underlying disease. This is in correspondence with previous studies showing that the compliance to follow-up is lower among older individuals and individuals with underlying disease [33]. In line with this, we observed a slightly lower compliance to follow-up in the oldest age group, during incidence screening the compliance to follow-up was 88.8% among this age group (Supplementary Table 6). Moreover, previous studies have shown a lower compliance to follow-up among immigrants, individuals living alone and individuals with a low income [27,28]. It is therefore important to understand barriers to compliance to follow-up among these groups of individuals to ensure adequate follow-up among all FIT-positive screening participants. Possibly, easier to understand information material or graphical information is needed. Moreover, for some individuals, there may be a need for a consultation with a healthcare professional before making the decision to undergo colonoscopy.

ADR is an important short-term marker of the quality of colonoscopies. We found that the ADR increased slightly over time and that there was a shift towards a higher proportion of low-risk adenomas. A study from the Netherlands, found a median ADR of 67% [34]; however, the FIT positivity threshold in that study was slightly lower (15 μg Hgb/g faeces) than in Denmark and the results may therefore not be directly comparable. Our study showed that the 3-year PCCRC rate using the WEO method [24] fluctuated around 7.1–8.9 during the study period and decreased slightly to 6.8 in 2021. The results are thus in line with a systematic review and meta-analysis by Kang et al reporting a pooled 3-year PCCRC of 8.2% (95% CI: 6.9–9.4, $I^2 = 98.2\%$) across four studies using WEO precise methodology [35]. The slight decline in PCCRC-rate in our study could indicate an increased quality of the colonoscopies performed. Moreover, it may reflect that training has been initiated to increase the performance of the colonoscopists and furthermore, it may reflect increased experience in performing colonoscopies over time [36].

We found that the number of CRC per 100,000 participants was high, though stable over time during prevalence screening, whilst the number of CRC per 100,000 participants decreased during incidence screening. This observation is as expected, since the aim of CRC is both detection of CRC at an earlier stage and removal of adenomas to prevent CRC from developing. At an aggregated level, we found an increase in the incidence of CRC in the entire population after the introduction of screening followed by a decline in the incidence over time. This observation is in line with previous studies [4,8,37]. There was a slight decline in the mortality of CRC in the entire population already before the introduction of screening, which continued after the introduction of screening. This is in accordance with some previous studies [11–13], though not all [9,10]. Possibly the implementation of CRC screening in Denmark in 2014 has not yet had an impact on the mortality of CRC given the limited follow-up time.

The proportion of positive FIT was 6.9% during prevalence screening and 4.7% during incidence screening. These estimates are largely in accordance with the findings of the systematic review by Baile-Maxía et al showing a pooled FIT positivity of 5.9% (6.6% 1st round, 5.6% 2nd round, 5.3% 3rd round, 5.3% 4th round) [25]. However, FIT positivity depends on the chosen threshold for positivity, and the FIT positivity threshold varies greatly between countries [38] thus results are difficult to compare. The FIT positivity thresholds also vary between the Nordic countries: in Norway, the FIT positivity threshold is 15 μg Hgb/g faeces, in Denmark 20 μg Hgb/g, in Finland 25 μg Hgb/g and in Sweden, the FIT positivity threshold is 40 μg Hgb/g for women and 80 μg Hgb/g for men [17]. As in Sweden, difference in cut-off value between women and men could also be considered in the CRC screening programme in Denmark, since using the same cut-off value results in a higher recall among men than women (5.5% vs 4.0% during incidence screening), and concurrently, the number of CRC per 100,000 invitees and per 100,000 participants was higher among men than among women. Nonetheless, these findings are based on a lower participation in CRC screening among men (65.9% vs 57.5% in women vs men during prevalence screening and 65.2% vs 55.7% during incidence screening). The true proportion of positive FIT among men (if they had the same participation as women) is therefore unknown. A lower participation in FIT-based screening and a higher CRC risk among men was also observed in a study from the Netherlands [39]. Nonetheless, the higher FIT positivity and higher CRC risk among men participating in screening could indicate a need for a more tailored or personalised approach to CRC screening taking into account information on e.g., age, sex, genetic factors and lifestyle factors.

4.3. Strengths and limitations

The strengths of the study comprise that it is based on nation-wide data from population-based registries, which have a high degree of

completeness and validity [15]. The study thus captures all data on invitations for CRC screening, participation, positive FIT, compliance to follow-up and adenomas and CRC in Denmark throughout the study period based on linkage of the data using the unique personal identifier. The study period includes the era of the COVID-19 pandemic; however, in Denmark, the FIT-based CRC screening programme continued throughout the pandemic as described in our previous publication (Olesen et al eLife 2023). A slight reduction in participation was observed at the start of the pandemic followed by slightly higher participation than before the pandemic. A slight reduction in compliance to colonoscopy was also observed at the start of the pandemic [40]. Limitations of the study comprise that measures of incidence and mortality is merely examined at the population-level and the direct impact of CRC screening in Denmark on the incidence and mortality therefore cannot be inferred.

5. Conclusions

The performance of the FIT-based CRC screening programme in Denmark is within the acceptable levels defined by the European guidelines. ADR increased slightly over time. The number of screen-detected CRC decreased over time. In the entire population, an initial increase in the incidence of CRC was observed followed by a decrease over time, while CRC mortality continued to decline. Continuous monitoring of the CRC screening programme in Denmark is warranted. Moreover, interventions are needed to increase participation and possibly a more tailored or personalised approach to CRC screening could be considered.

CRediT authorship contribution statement

Mikkel Sjöström: Writing – review & editing, Validation, Methodology, Investigation. **Bo Søndergaard:** Writing – review & editing, Validation, Methodology, Investigation. **Morten Rasmussen:** Writing – review & editing, Validation, Methodology, Investigation. **Tina Bech Olesen:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Berit Andersen:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Signe Bülow Therkildsen:** Writing – review & editing, Formal analysis, Conceptualization. **Mette Høyrup:** Writing – review & editing. **Henry Jensen:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Sanne Kjær-Frifeldt:** Writing – review & editing, Validation, Methodology, Investigation. **Lasse Pedersen:** Writing – review & editing, Validation, Methodology, Investigation. **Petra Hall Viborg:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Nina Brander:** Writing – review & editing, Validation, Methodology, Investigation. **Lennart Friis-Hansen:** Writing – review & editing, Validation, Methodology, Investigation.

Funding statement

No specific funding was assigned for this study.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

- Dr Morten Rasmussen declares lecture honoraria from Olympus and Norgine.
- Professor Berit Andersen declares the following: Head of CRC screening in Central Denmark Region and member of the national steering committees for CRC screening in Denmark.
- The remaining authors state that they have nothing to declare.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ejca.2026.116788.

References

[1] GLOBOCAN Global Cancer Observatory [Internet]; [cited 24-02-2025]. Available from: ⟨https://gco.iarc.fr/today/en/dataviz/tables?mode=cancer&group_populations=1&multiple_populations=1⟩.
[2] International Agency for Research on Cancer (IARC). NORDCAN 2025; [updated 07-10-2025]. Available from: ⟨https://nordcan.iarc.fr/en⟩.
[3] von Karsa LA A, Ronco G, Ponti A, Malila N, Arbyn M, Segnan N, Castillo-Beltran M, Boniol M, Ferlay J, Hery C, Sauvaget C, Voti L, Autier P. Cancer screening in the European Union – report on the implementation of the Council Recommendation on cancer screening – first report. International Agency for Research on Cancer (IARC); 2008.
[4] Cardoso R, Guo F, Heisser T, Hackl M, Ihle P, De Schutter H, et al. Colorectal cancer incidence, mortality, and stage distribution in European countries in the colorectal cancer screening era: an international population-based study. Lancet Oncol 2021; 22(7):1002–13.
[5] Dekker E, Tanis PJ, Vleugels JLA, Kasi PM, Wallace MB. Colorectal cancer. Lancet 2019;394(10207):1467–80.
[6] Lansdorp-Vogelaar I, von, Karsa L. European guidelines for quality assurance in colorectal cancer screening and diagnosis. First Edition–Introduction. Endoscopy 2012;44 3:Se15–30.
[7] European Commission Initiative on Colorectal Cancer (ECICC). European guidelines on colorectal cancer screening and diagnosis; 2025 [Available from: ⟨https://cancer-screening-and-care.jrc.ec.europa.eu/en/ecicc/european-colorectal-cancer-guidelines?topic=289&usertype=282⟩].
[8] Carbotti G, van den Berg DMN, Carvalho AL, Senore C, Heijnsdijk EAM, de Koning HJ, et al. Developments in the roll-out and performance of CRC screening in Europe. Best Pract Res Clin Gastroenterol 2025.
[9] Njor SH, Larsen MB, Søborg B, Andersen B. Colorectal cancer mortality after randomized implementation of FIT-based screening – a nationwide cohort study. J Med Screen 2022;29(4):241–8.
[10] Zheng S, Chen K, Liu X, Ma X, Yu H, Chen K, et al. Cluster randomization trial of sequence mass screening for colorectal cancer. Dis Colon Rectum 2003;46(1):51–8.
[11] Gini A, Jansen EEL, Zielonke N, Meester RGS, Senore C, Anttila A, et al. Impact of colorectal cancer screening on cancer-specific mortality in Europe: a systematic review. Eur J Cancer 2020;127:224–35.
[12] Jodal HC, Helsingen LM, Anderson JC, Lytvyn L, Vandvik PO, Emilsson L. Colorectal cancer screening with faecal testing, sigmoidoscopy or colonoscopy: a systematic review and network meta-analysis. BMJ Open 2019;9(10):e032773.
[13] Fitzpatrick-Lewis D, Ali MU, Warren R, Kenny M, Sherifali D, Raina P. Screening for colorectal cancer: a systematic review and meta-analysis. Clin Colorectal Cancer 2016;15(4):298–313.
[14] Population [Internet]; 2025 [cited 30-01-2025]. Available from: ⟨https://www.dst.dk/en/Statistik/emner/borgere/befolkning/befolkningstal⟩.
[15] Schmidt M, Schmidt SAJ, Adelborg K, Sundbøll J, Laugesen K, Ehrenstein V, et al. The Danish health care system and epidemiological research: from health care contacts to database records. Clin Epidemiol 2019;11:563–91.
[16] Thomsen MK, Njor SH, Rasmussen M, Linnemann D, Andersen B, Baatrup G, et al. Validity of data in the Danish Colorectal Cancer Screening Database. Clin Epidemiol 2017;9:105–11.
[17] Jorgensen SF, Njor SH, Nevala A, Palsson B, Randel KR, Agustsson AI, et al. Nordic colorectal cancer screening programmes: a comparison of organization, operation, and quality indicators. Eur J Cancer 2025;222:115444.
[18] Hewitson P, Glasziou P, Watson E, Towler B, Irwig L. Cochrane systematic review of colorectal cancer screening using the fecal occult blood test (hemoccult): an update. Am J Gastroenterol 2008;103(6):1541–9.
[19] Garborg K, Holme Ø, Løberg M, Kalager M, Adami HO, Bretthauer M. Current status of screening for colorectal cancer. Ann Oncol 2013;24(8):1963–72.
[20] Selskab DP. Screenings- og adenomkontrol program for tyk- og endetarmskræft - Guidelines for koloskopi og patologi. 2014.
[21] Sundhedsstyrelsen. Anbefalinger vedrørende screening for tyk- og endetarmskræft. 2012.
[22] Schmidt M, Schmidt SA, Sandegaard JL, Ehrenstein V, Pedersen L, Sorensen HT. The Danish National Patient Registry: a review of content, data quality, and research potential. Clin Epidemiol 2015;7:449–90.
[23] Schmidt M, Pedersen L, Sørensen HT. The Danish civil registration system as a tool in epidemiology. Eur J Epidemiol 2014;29(8):541–9.
[24] Rutter MD, Beintaris I, Valori R, Chiu HM, Corley DA, Cuatrecasas M, et al. World endoscopy organization consensus statements on post-colonoscopy and post-imaging colorectal cancer. Gastroenterology 2018;155(3):909–25. e3.
[25] Baile-Maxía S, Pimienta M, Sánchez-Ardila C, Castells A, Jover R, Ladabaum U. Systematic review of participation, positivity, and yield over time in fecal immunochemical test-based organized colorectal cancer screening programs. Clin Gastroenterol Hepatol 2026.
[26] Larsen MB, Mikkelsen EM, Rasmussen M, Friis-Hansen L, Ovesen AU, Rahr HB, et al. Sociodemographic characteristics of nonparticipants in the Danish colorectal cancer screening program: a nationwide cross-sectional study. Clin Epidemiol 2017;9:345–54.

[27] Pallesen AVJ, Herrstedt J, Westendorp RGJ, Mortensen LH, Kristiansen M. Differential effects of colorectal cancer screening across sociodemographic groups in Denmark: a register-based study. Acta Oncol 2021;60(3):323–32.
[28] Olesen TB, Jensen H, Moller H, Jensen JW, Andersen B, Rasmussen M. Nationwide participation in FIT-based colorectal cancer screening in Denmark during the COVID-19 pandemic: an observational study. Elife 2023;12.
[29] Society TDC. Campaign: "Hvornår har du sidst synet dig selv?"; 2025 [Available from: ⟨https://www.cancer.dk/forebyg-kraeft/screening/tarmkraeft/syn-dig-selv/⟩.
[30] Larsen MB, Hedelund M, Flander L, Andersen B. The impact of pre-notifications and reminders on participation in colorectal cancer screening – a randomised controlled trial. Prev Med 2022;164:107229.
[31] Robb KA, Young B, Murphy MK, Duklas P, McConnachie A, Hollands GJ, et al. Behavioural interventions to increase uptake of FIT colorectal screening in Scotland (TEMPO): a nationwide, eight-arm, factorial, randomised controlled trial. Lancet 2025;405(10484):1081–92.
[32] Helgestad ADL, Larsen MB, Njor S, Tranberg M, Petersen LK, Andersen B. Increasing coverage in cervical and colorectal cancer screening by leveraging attendance at breast cancer screening: a cluster-randomised, crossover trial. PLoS Med 2024;21(8):e1004431.
[33] Thomsen MK, Rasmussen M, Njor SH, Mikkelsen EM. Demographic and comorbidity predictors of adherence to diagnostic colonoscopy in the Danish Colorectal Cancer Screening Program: a nationwide cross-sectional study. Clin Epidemiol 2018;10:1733–42.
[34] Wisse PHA, Erler NS, de Boer SY, den Hartog B, Oudkerk Pool M, Terhaar Sive Droste JS, et al. Adenoma detection rate and risk for interval postcolonoscopy colorectal cancer in fecal immunochemical test-based screening: a population-based cohort study. Ann Intern Med 2022;175(10):1366–73.
[35] Kang JH, Evans N, Singh S, Samadder NJ, Lee JK. Systematic review with meta-analysis: the prevalence of post-colonoscopy colorectal cancers using the World Endoscopy Organization nomenclature. Aliment Pharm Ther 2021;54(10):1232–42.
[36] Pedersen L, Bernstein I, Lindorff-Larsen K, Carlsen CG, Torp-Pedersen C. Improving colonoscopy quality through individualised training programmes. Dan Med J 2020;67(8).
[37] Johannsen IR, Boysen AK, Mortensen FV, Kirkegard J. Temporal trends in incidence and mortality of colorectal cancer in Denmark from 2007 to 2022. Int J Cancer 2025;157(4):634–43.
[38] Young GP, Benton SC, Bresalier RS, Chiu HM, Dekker E, Fraser CG, et al. Fecal immunochemical test positivity thresholds: an international survey of population-based screening programs. Dig Dis Sci 2025;70(5):1637–45.
[39] van de Schootbrugge-Vandermeer HJ, de Jonge L, van Leerdam ME, van Vuuren AJ, Dekker E, Spaander MCW, et al. Risk of colorectal cancer and participation in fecal immunochemical test-based screening. Public Health 2025;248:105938.
[40] Olesen TB, Jensen H, Møller H, Jensen JW, Andersen B, Rasmussen M. Nationwide participation in FIT-based colorectal cancer screening in Denmark during the COVID-19 pandemic: an observational study. Elife 2023;12.